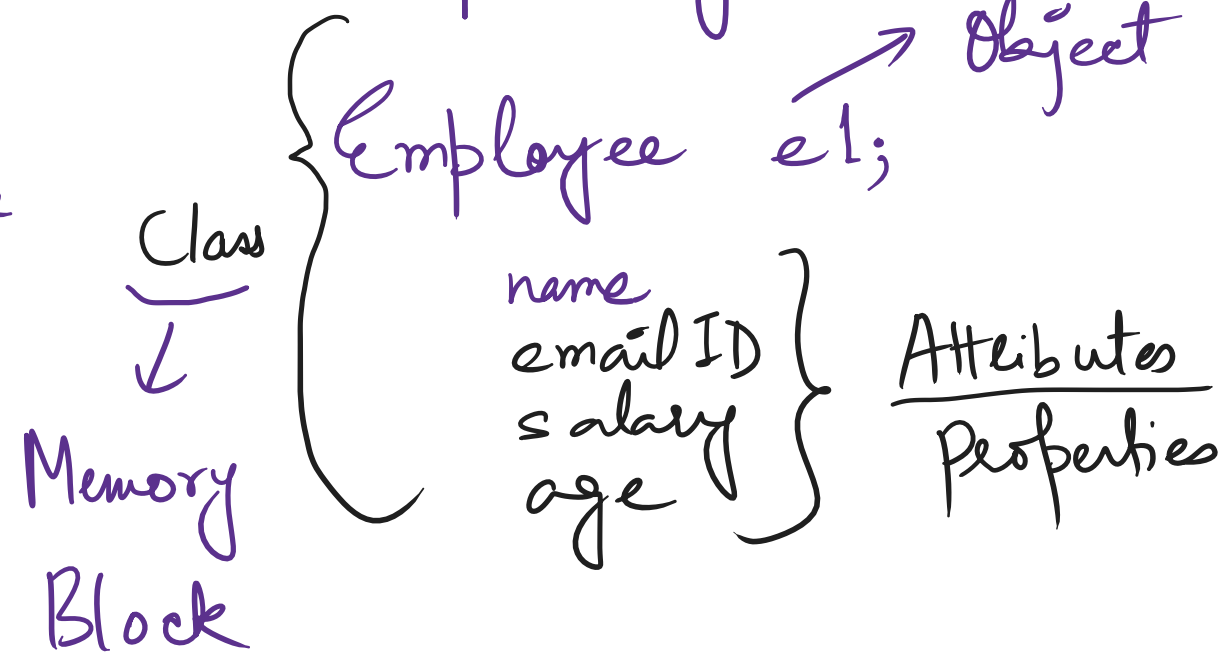


Object Oriented Programming →

* Solving real world problems, and representing real world entities.

- * Class
- * Object / Instance / Reference
- * Attributes / Properties
- * Methods
- * Code Reusability



Class → It is a blueprint to create objects / references / instances. It has no memory of its own (**). It is just used to instantiate objects.

Object → It is an instance of a class. It is a key by which we can access all the properties (data members) inside the class. It is also called a reference.

Methods → These are functions that are created inside a class. They are called methods because they describe the behaviour of an object.

Attributes → These are the common properties / parameters that are assigned to each object of the class.

Example: (Employee) → name, age, salary, etc.

Constructors → These are special methods which are used to instantiate objects.

* Rules to create constructors:

- Same name as the class followed by () Parentheses.
- It has no return type.
- It has to be public or else we cannot create objects.
- There are two important types:

- ① **Default Constructor**: It is generated by the compiler if we don't create it.
- ② **Parameterized Constructor**: It is created by the user.
- No-argument constructor.*

Name Clash: Name clashes between class attributes can be resolved in two ways:

① Writing different variable names:

② Writing same names & using 'this' pointer.

```
// Attributes // Properties
string Name;
string emailID;
double salary;
string phoneNumber;
// Constructor
Employee(string name, string email, double s, string phone){
    Name = name;
    emailID = email;
    salary = s;
    phoneNumber = phone;
}
```

```
// Attributes // Properties
string Name;
string emailID;
double salary;
string phoneNumber;
// Constructor
Employee(string Name, string emailID, double salary, string phoneNumber){
    this->Name = Name;
    this->emailID = emailID;
    this->salary = salary;
    this->phoneNumber = phoneNumber;
}
```

- * Pillars:
- ① Inheritance
 - ② Encapsulation
 - ③ Polymorphism
 - ④ Abstraction

Book

- title
- ISBN
- author
- price

bookDetails();

* Normal

* Constructor